

Assignment 3

Q1) Explain the basic concept of a Decision Tree. What are its main components and how is it used in classification tasks?

Basic Concept of a Decision Tree:

A **Decision Tree** is a supervised machine learning model used for both **classification** and **regression** tasks. It resembles a **flowchart-like structure**, where each **node represents a decision** based on a feature, and the **paths (branches)** lead to different outcomes. It works by **recursively splitting the dataset** based on feature values to form a tree that leads to predictions.

Main Components of a Decision Tree:

1. **Root Node:**
 - Represents the **entire dataset**.
 - Splitting starts here using the **most informative feature**.
2. **Internal (Decision) Nodes:**
 - Each node performs a **test on a feature** (e.g., "Is age > 30?").
 - Based on the result, the data is split further.
3. **Branches:**
 - Represent the **outcome of a decision** or test.
 - Connect a node to its child nodes.
4. **Leaf (Terminal) Nodes:**
 - Represent the **final output** (e.g., a class label).
 - No further splits occur at this point.

How It's Used in Classification Tasks:

1. Training Phase:

- The algorithm:
 - Chooses the **best features** to split the data.
 - Uses metrics like:
 - **Gini Impurity**
 - **Entropy** (Information Gain)
 - Recursively splits the data to form a tree.
 - Stops splitting when:
 - Maximum depth is reached
 - A node is pure (all samples belong to one class)
 - Minimum samples per leaf is met

2. Prediction Phase:

- A new instance is passed through the tree:
 - At each node, the feature condition is evaluated.
 - The instance follows the branch that satisfies the condition.
 - This continues until a **leaf node** is reached.
 - The class label of the leaf is the **predicted output**.

Example:

If the decision tree is built to classify whether a person will buy a product:

- Root Node: "Is Age > 30?"
- If Yes → "Is Income > 50k?"
- If Yes → **Buy = Yes** (Leaf)
- If No → **Buy = No** (Leaf)
- If Age ≤ 30 → **Buy = No** (Leaf)

Q2) Describe the purpose of attribute selection measures in Decision Trees. Compare Information Gain, Gain Ratio, and Gini Index with suitable examples.

Purpose of Attribute Selection Measures in Decision Trees:

Attribute selection measures are **used to decide which feature (attribute)** should be used to split the dataset at each node in a Decision Tree. The goal is to **select the attribute that best separates the data** into distinct classes, resulting in a more accurate and efficient tree.

Common Attribute Selection Measures:

Measure	Purpose	Best Used When
Information Gain (IG)	Measures the reduction in entropy after a split.	When dataset has balanced attributes.
Gain Ratio	Adjusts IG by accounting for attribute bias.	When attributes have many distinct values.
Gini Index	Measures impurity; lower Gini means better split.	Used in CART (Classification & Regression Trees).

1. Information Gain (IG):

- **Definition:** $IG = Entropy(\text{before split}) - \text{Weighted Entropy}(\text{after split})$
- **Goal:** Choose the attribute that gives the **maximum IG**.
- **Bias:** Prefers attributes with many values.

Example:

Suppose we want to classify people as "Buys Computer" based on "Age".

- Before split: Entropy = 0.94
- After split by "Age": Entropy = 0.69
- **Information Gain = $0.94 - 0.69 = 0.25$**

2. Gain Ratio:

- **Definition:** $\text{Gain Ratio} = \text{Information Gain} / \text{Split Information}$
- **Purpose:** Solves IG's bias toward attributes with many values.
- **Goal:** Choose attribute with **highest Gain Ratio**.

Example:

- Attribute A1: $IG = 0.5, \text{SplitInfo} = 0.9 \rightarrow \text{Gain Ratio} = 0.56$
- Attribute A2: $IG = 0.6, \text{SplitInfo} = 1.5 \rightarrow \text{Gain Ratio} = 0.4$
- ☒ A1 is chosen (higher gain ratio)

3. Gini Index:

- **Definition:** $Gini(D) = 1 - \sum (p_i)^2$, where p_i is the probability of class i .
- **Goal:** Choose attribute that results in **lowest Gini** after split.
- **Faster to compute** than entropy.

Example:

If a dataset has 60% Yes and 40% No:

- $Gini = 1 - (0.6)^2 - (0.4)^2 = 0.48$
- Split the dataset and compute Gini for each subset. Choose the attribute with **lowest combined Gini**.

Comparison Summary:

Measure	Handles Many Values?	Biased?	Based On	Best For
Info Gain	☒ No	☒ Yes	Entropy	Simple, balanced attributes
Gain Ratio	☒ Yes	☒ No	Adjusted Entropy	Attributes with many values
Gini Index	☒ Yes	☒ Less	Impurity	CART, fast computation

Scenario Based Question

Q1) A hospital is developing an Intelligent System to assess the severity of a patient's flu symptoms using fuzzy logic. The system considers three input factors such as body temperature, cough severity and fatigue. For this purpose, explain how fuzzy logic helps the system decide whether a patient needs home rest, a doctor visit or immediate hospitalization. A popular ride-sharing company, Swift-wheels, introduces a new AI- powered system for dynamic pricing. This system adjusts ride fares in real-time based on factors like demand, traffic, time of day, and even the user's past ride history. It's a rainy evening, and many people are trying to get home. Sarah, who lives in a lower-income neighborhood, tries to book a Swift Wheels ride. From this purpose Explain From an AI ethics perspective, what is the core ethical dilemma presented in this scenario? Briefly explain why it's a dilemma and suggest one way Swift-wheels could try to address it.

Part 1: Fuzzy Logic in Hospital’s Intelligent System

Fuzzy logic is useful in handling **imprecise, uncertain, or vague data**, which is common in medical assessments. Here's how it applies:

Inputs:

1. **Body Temperature** (e.g., Low, Normal, High, Very High)
2. **Cough Severity** (e.g., Mild, Moderate, Severe)
3. **Fatigue** (e.g., Low, Medium, High)

Process:

- Fuzzy logic converts **crisp inputs** into **fuzzy values** using **membership functions**.
- It applies a set of **IF-THEN rules** like:
 - *IF* temperature is High **AND** cough is Severe **AND** fatigue is High → *THEN* Immediate Hospitalization
 - *IF* temperature is Mild **AND** cough is Moderate → *THEN* Doctor Visit
 - *IF* all symptoms are Low → *THEN* Home Rest
- After applying rules, the system performs **defuzzification** to convert fuzzy output into a clear decision.

Output:

- Recommendation: **Home Rest, Doctor Visit, or Immediate Hospitalization**

Fuzzy logic allows the system to mimic human reasoning when dealing with vague medical symptoms.

Part 2: AI Ethics in Swift-Wheels Dynamic Pricing

Core Ethical Dilemma:

Algorithmic Bias and Fairness

The AI system raises prices during high demand (e.g., rain), but **Sarah**, who lives in a **lower-income area**, faces higher fares, possibly **pricing her out** of access to essential transport.

Why It's a Dilemma:

- The system is **economically efficient** but risks **discriminating** against people in low-income areas or with certain ride histories.
- It unintentionally **amplifies inequality** under the guise of dynamic pricing.

Suggested Solution:

Implement **Fairness Constraints** in the pricing model.

Swift-Wheels could:

- Set **upper limits** on price hikes in economically disadvantaged areas.
- Introduce **subsidies or discounts** during emergencies or bad weather.
- Audit the AI system regularly for **disparate impacts** on different socioeconomic groups.

Q2) Explain the concept of a Decision Tree, its main components and how is it used in classification tasks? Identify a major tech company develop a new AI-powered facial recognition system that can identify individuals with near-perfect accuracy, even in challenging conditions. Explain the key ethical considerations and emerging AI trends must the company prioritize and address before widely deploying this powerful technology?

Part 1: Decision Tree — Concept, Components, and Use in Classification

Concept:

A **Decision Tree** is a **supervised machine learning algorithm** used for **classification and regression**. It splits data into branches based on feature values, forming a tree-like structure that leads to a decision or prediction.

Main Components:

1. **Root Node**
 - Represents the entire dataset.
 - Splits into branches based on the most significant feature.
2. **Internal Nodes (Decision Nodes)**
 - Represent decisions/tests on features (e.g., *Is Age > 50?*).
 - Lead to further splits.
3. **Branches**
 - Outcomes of a decision, guiding to the next node.
4. **Leaf Nodes (Terminal Nodes)**
 - Final output/class (e.g., *Yes, No, Class A*).

Used in Classification Tasks:

- For example, classifying whether an email is spam:
 1. Input features: keywords, sender, number of links.
 2. The tree splits based on these features.

3. A leaf node gives the final classification (Spam/Not Spam).

Part 2: Ethical Considerations in AI-Powered Facial Recognition

Example Company: Apple (or Google, Meta)

Suppose Apple develops a **highly accurate facial recognition system**.

Key Ethical Considerations:

1. **Privacy Invasion**
 - Facial data is biometric and sensitive.
 - Risk of mass surveillance or misuse without consent.
2. **Bias & Discrimination**
 - Historical datasets may introduce **racial, gender, or age biases**, leading to unfair results.
3. **Consent and Transparency**
 - Individuals must **know and agree** that their face is being scanned.
4. **Misuse by Governments or Third Parties**
 - Risk of it being used for surveillance, oppression, or targeting minority groups.
5. **Security Risks**
 - If facial data is leaked, it's **irreplaceable**, unlike passwords.

Emerging AI Trends to Address:

1. **Explainable AI (XAI)**
 - Make facial recognition decisions **understandable** to users and regulators.
2. **Federated Learning**
 - Training models on-device to **reduce data sharing** and enhance privacy.
3. **Differential Privacy**
 - Add noise to data to preserve individual identity while still training the model.
4. **Ethical AI Auditing**
 - Regular evaluation for **fairness, bias, and compliance** with regulations.

Suggestion:

Before deployment, the company must conduct **impact assessments**, ensure **regulatory compliance (like GDPR)**, and **involve ethicists and diverse stakeholders** in the design process.